POWER PULSE

Measurement of Voltage Important Formulae

Notes by BALAJI SIR

1 DC voltmeter with multiplier

1. $$m = \frac{V}{V_m}, \qquad R_s = (m-1)R_m = \frac{V}{I_m} - R_m$$

V_m = full-scale meter voltage; R_s = series multiplier

2. $$S = \frac{1}{I_{fsd}}\ \Omega/\mathrm{V}, \qquad R_v = S \times V_{range}$$

voltmeter sensitivity; a high S means little loading

2 Loading effect

3. $$V_{read} = V_{oc}\,\frac{R_v}{R_v + R_{th}}$$

R_th = Thevenin resistance seen by the voltmeter; V_oc = open-circuit voltage

4. $$\%\text{error} = -\frac{R_{th}}{R_{th} + R_v} \times 100$$

3 AC voltmeter

5. $V_{rms} = 1.11\,V_{avg}$ (rectifier, sine), peak-reading: $V_{rms} = 0.707\,V_p$

the scale is calibrated for a sine wave; correct the reading for other wave shapes

4 Potential transformer

6. $$V_{line} = K_n\,V_s \quad (V_s = 110\ \mathrm{V\ nominal})$$

5 DC potentiometer

7. $$E_x = E_s\,\frac{l_x}{l_s}$$

draws no current at balance, so there is no loading error

6 Electrostatic voltmeter

8. $$T_d = \frac{1}{2}V^2\frac{dC}{d\theta}$$

square law; reads rms; draws no current; used for high voltages